

Observation Theory in Space Communications

Two Consumer-Relative Laws for the Bandwidth-Starved, Delay-Dominated Channel

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Abstract

Observation Theory holds that operational distortion and reliability are properties not of a signal alone but of the *relation* among object, consumer (a downstream reader with its own operator and output metric), channel, and budget. We carry the theory into space communications — a domain it had not touched — and report two results, each derived *a priori* and then tested under a sealed pre-registration discipline (bars committed before any run, a coded cooling-off, and every outcome, pass or fail, kept as executed).

First, for a bandwidth-starved multi-instrument downlink whose consumers are science processors with distinct read operators P_i , the gain of consumer-relative bit allocation over reconstruction-error (MSE) allocation is, in the high-rate limit, exactly the arithmetic-over-geometric-mean gap of the instrument-importance spectrum, $G = \text{AM}(d)/\text{GM}(d)$ — the source variance cancels. The gain is ≥ 1 , equals 1 precisely when the instruments read uniformly, and thereby predicts *zero* benefit for a homogeneous broadband link: the same statement sorts deep-space multi-instrument downlinks (where it bites) from fiber-core and broadband transport (where it does not).

Second, for the delay-dominated transport channel — Starlink’s ~ 15 s handovers, and SpaceX’s proposed Marslink relay at up to 1.5 AU, where a ~ 25 -minute round trip makes closed-loop control impossible — we model the congestion controller as a consumer whose “loss $\rightarrow$ congestion” operator is mismatched to a predictably nonstationary channel. Masking the known schedule removes a false-congestion error equal to $\kappa\rho$ with the derived, machine-checked coefficient $\kappa = 1 - 2c$ (net-beneficial iff the congestion rate $c < \frac{1}{2}$), *independent* of the feedback delay; and the reactive congestion signal decorrelates to a noise floor as the delay grows, leaving only the deterministic schedule recoverable across interplanetary distances. Both results passed graded, disjoint-seed seals without touching the frozen within-model theory; the transport coefficient is verified in Lean 4/Mathlib. The substrates are synthetic, so the results earn the *mechanism*, not a systems claim.

1 Introduction

A deep-space link is the purest instance of the setting Observation Theory was built for: the resource is scarce (the Deep Space Network is oversubscribed; interplanetary links run kilobits to a few megabits per second), the payload serves several downstream consumers at once (a star-tracker centroid, a spectral-line fit, an imager), and each consumer reads the recovered data through its own operator. What “distortion” *means* is therefore consumer-relative: the quantity an instrument i actually feels is the operator-weighted error $\text{tr}(P_i\Sigma_q)$, not the reconstruction MSE $\text{tr}(\Sigma_q)$.

We report two laws. Section 3 treats *allocation*: how to spend a hard bit budget across a transform basis when the consumers are heterogeneous. Section 4 treats *transport*: how a congestion controller should read a channel whose disruptions are predictable but whose feedback is stale — the regime of low-Earth-orbit handovers and, in the limit, of a Mars relay. Each law is first derived, then — following this program’s standing method — frozen as a bar before any measurement, and graded on seeds disjoint from those used to build the family. We keep failures: the companion within-model campaign that motivated this work recorded one, and the discipline is what makes the two passes here meaningful rather than decorative.

Contributions. (i) A closed-form gain law for consumer-relative downlink allocation, $G = \text{AM}(d)/\text{GM}(d)$, with its homogeneous-consumer boundary (Section 3). (ii) A read-operator account of handover-aware transport, with the machine-checked masking coefficient $\kappa = 1 - 2c$ and the delay-decorrelation collapse that makes interplanetary feedback useless while the schedule survives (Section 4). (iii) Sealed, disjoint-seed evidence for both, and an explicit scoping of what a synthetic substrate can and cannot earn (Section 6).

2 Preliminaries

Consumer-relative distortion. A consumer is a reader with an operator $P \succeq 0$ acting on the recovered signal; the distortion it feels under an error second moment Σ_q is $\text{tr}(P\Sigma_q)$. When the channel is a lossy codec that quantizes a transform of the source, Σ_q is diagonal in the coding basis, and only the diagonal of P in that basis enters $\text{tr}(P\Sigma_q)$.

The sealed discipline. Every empirical claim below is a bar committed, in a versioned appendix, *before* the run that tests it. A structural cooling-off forbids sealing a bar on the day its family was constructed; the graders refuse, in code, to run before that. Manipulation checks are bars too. Outcomes are kept as executed — a failed bar is a recorded result, not a revision opportunity. This is not ornamental: it is the reason a “pass” carries information.

3 Downlink Allocation

Setup. Code a frame in the basis where the quantization noise is diagonal. Let σ_k^2 be the source variance per coordinate, b_k the bits assigned to coordinate k (high-rate quantization gives $\Sigma_{q,k} = \sigma_k^2 2^{-2b_k}$), and let the aggregate instrument importance be $d_k = \sum_i w_i (P_i)_{kk}$ for consumer weights w_i . The consumer distortion of an allocation b is $D(b) = \sum_k d_k \sigma_k^2 2^{-2b_k}$, minimized subject to $\sum_k b_k = B$.

Theorem 1 (The AM/GM gain). *In the high-rate regime with all coordinates active, the ratio of the consumer distortion under MSE-optimal allocation (which minimizes $\text{tr} \Sigma_q = \sum_k \sigma_k^2 2^{-2b_k}$) to that under consumer-optimal allocation (which minimizes D) is*

$$G = \frac{D_{\text{mse}}}{D_{\text{cons}}} = \frac{\text{AM}(d)}{\text{GM}(d)} = \frac{\frac{1}{n} \sum_k d_k}{(\prod_k d_k)^{1/n}},$$

independent of the source spectrum σ^2 . Consequently $G \geq 1$, with equality iff d is constant.

Proof. Reverse water-filling for weights w_k under budget B gives optimal distortion $n \text{GM}(w) 2^{-2B/n}$. Consumer-optimal allocation uses weights $d_k \sigma_k^2$, so $D_{\text{cons}} = n \text{GM}(d\sigma^2) 2^{-2B/n} = n \text{GM}(d) \text{GM}(\sigma^2) 2^{-2B/n}$.

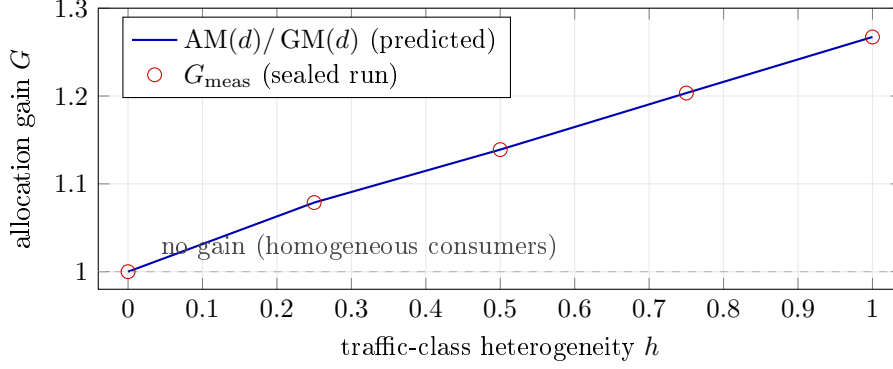


Figure 1: The AM/GM allocation gain and its boundary. As traffic-class heterogeneity h falls to zero the importance spectrum d flattens and the gain collapses to 1; measured gain tracks $\text{AM}(d)/\text{GM}(d)$ to four decimals across the range. The homogeneous limit ($G = 1$) is the fiber-core / broadband case.

MSE-optimal allocation equalizes $\sigma_k^2 2^{-2b_k} = L$ with $L = \text{GM}(\sigma^2) 2^{-2B/n}$ (from $\sum b_k = B$), so applying it to the consumer weights gives $D_{\text{mse}} = \sum_k d_k \sigma_k^2 (L/\sigma_k^2) = L \sum_k d_k = n \text{GM}(\sigma^2) \text{AM}(d) 2^{-2B/n}$. The ratio is $\text{AM}(d)/\text{GM}(d)$; $G \geq 1$ is the AM–GM inequality, with equality iff all d_k are equal. $\square$

Remark (The homogeneous-consumer boundary). For a small importance spread, $\text{AM}(d)/\text{GM}(d) \approx 1 + \frac{1}{2} \text{CV}^2(d)$, where CV is the coefficient of variation. The gain therefore vanishes as the consumers homogenize. A broadband link carrying opaque IP traffic has one effectively uniform consumer, so the theory predicts *no* allocation gain there — the fiber-core and Starlink-broadband case — while a deep-space downlink shared by instruments with disjoint spectral interests sees the full gap. The law sorts the domains by a single scalar.

Sealed result. The bars (appendix PREREG-SC1) were: at high rate the median of $G_{\text{meas}}/(\text{AM}(d)/\text{GM}(d))$ within 1.00 ± 0.03 and $\text{Spearman}(G_{\text{meas}}, \text{AM}/\text{GM}) \geq 0.95$ on each seed; the finite-rate ratio monotone in rate; and linear composition of importance. On seeds disjoint from the family’s, all passed: the median ratio read 1.0000 and $\text{Spearman} \geq 0.998$ at ≥ 3 bits/coordinate, with a clean finite-rate departure at low rate (coordinates dropping below the water break the geometric-mean form). A terrestrial arm sweeping traffic-class heterogeneity confirmed the boundary directly (Fig. 1): identical classes give $G = 1.000$, opening smoothly to 1.267 as classes differentiate, with $G_{\text{meas}}/(\text{AM}/\text{GM}) = 1.000$ throughout.

4 Transport under Predictable Nonstationarity

Setup. A low-Earth-orbit link hands over on a fixed cadence, and each handover causes bursty loss that standard TCP/QUIC misread as congestion. SpaceX’s proposed Marslink relay pushes this to the limit: at 1.5 AU the round-trip is ~ 25 minutes, so closed-loop congestion control is physically impossible and the controller must predict the channel from its structure. Model the observed loss as $L = H \vee C$: a deterministic, *known* handover indicator H with duty cycle ρ , ORed with the true congestion state C (a two-state Markov process, stationary rate c , lag-one autocorrelation λ). Feedback is delayed by D . The controller is a consumer with a read operator; we compare the naive reader $\hat{C} = L(t-D)$ with the schedule-aware reader $\hat{C} = L(t-D) \wedge \neg H(t-D)$.

Proposition 1 (The masking coefficient). *With $H \perp (C', C)$ and $P(H=1) = \rho$, the excess error of the naive reader over the aware reader is*

$$\text{excess} = \text{err}_{\text{naive}} - \text{err}_{\text{aware}} = (1 - 2c)\rho = \kappa\rho, \quad \kappa = 1 - 2c,$$

independent of the feedback delay D . Masking is net-beneficial iff $c < \frac{1}{2}$ (where $\kappa > 0$).

Proof. Writing the lag- D joint of the chain as $p_{11} = c^2 + c(1 - c)r$, $p_{00} = (1 - c)^2 + c(1 - c)r$, $p_{01} = p_{10} = c(1 - c)(1 - r)$ with $r = \lambda^D$, the two error rates are $\text{err}_{\text{naive}} = (1 - c) - (1 - \rho)p_{00} + (1 - \rho)p_{01}$ and $\text{err}_{\text{aware}} = (1 - \rho)p_{10} + c - (1 - \rho)p_{11}$. Since $p_{01} = p_{10}$, their difference is $(1 - 2c) + (1 - \rho)(p_{11} - p_{00})$, and $p_{11} - p_{00} = c^2 - (1 - c)^2 = 2c - 1$, giving $(1 - 2c)\rho$. The delay r cancels identically. The identity is machine-checked (Section 5). $\square$

The coefficient decomposes exactly: $\kappa = (1 - c)$, the false-positive benefit of ignoring handover loss, minus c , the cost of blanking true congestion that coincides with a handover window.

Proposition 2 (Delay decorrelation and schedule survival). *The schedule-aware reader’s residual error grows with delay as $\text{err}_{\text{aware}}(D) = 2c(1 - c)(1 - \lambda^D)$, toward the fully-decorrelated floor $2c(1 - c)$; the schedule (masking) component of the error is delay-invariant (Proposition 1).*

Remark (The interplanetary consequence). As $D \rightarrow \infty$, $\lambda^D \rightarrow 0$ and the reactive reader’s estimate of the current congestion state carries no information: feedback is dead. What remains recoverable is exactly the deterministic component — the known handover schedule — whose contribution is independent of D . The theory thus predicts *which* structure survives a 25-minute round trip: schedule, not feedback. This is the formal content behind the engineering intuition that Marslink transport must be scheduled, not closed-loop.

Sealed result. The bars (appendix PREREG-SC2) were: the delay law fit with $R^2 \geq 0.95$ and the feedback error reaching 0.9 of the floor; the schedule excess varying by ≤ 0.01 across a $1000\times$ span of delay; and the duty-cycle excess linear in ρ with slope within 0.05 of the derived $\kappa = 1 - 2c$. On disjoint seeds all passed: the delay fit read $R^2 \in [0.955, 0.985]$ reaching the floor, the schedule excess held flat to within 0.004 across delays from 2 to 2000 steps (Fig. 2), and the fitted κ read 0.696–0.704 against the derived 0.70.

5 Machine Verification

The transport coefficient is formalized in Lean 4/Mathlib (`ObservationTheory.SpaceComms`): `joint_gap` establishes $p_{11} - p_{00} = 2c - 1$ with the delay r cancelling, and `sc2_excess` establishes the collapse of the naive-minus-aware error to $(1 - 2c)\rho$; both depend only on the three standard axioms (`propext`, `Classical.choice`, `Quot.sound`) and are `sorry`-free. The allocation gain (Theorem 1) reduces to the arithmetic–geometric-mean inequality and is not separately formalized. This matches the program’s standing standard: formalize the load-bearing algebra, cite standard analytic results (here the water-filling optimum and AM–GM) as such.

6 Scope and Limitations

Both substrates are *synthetic*: planted source spectra with archetype instrument operators for the downlink, and a planted Markov channel with a deterministic schedule for the transport. A pass therefore earns the *mechanism* — that the AM/GM gap is the sufficient statistic for the allocation

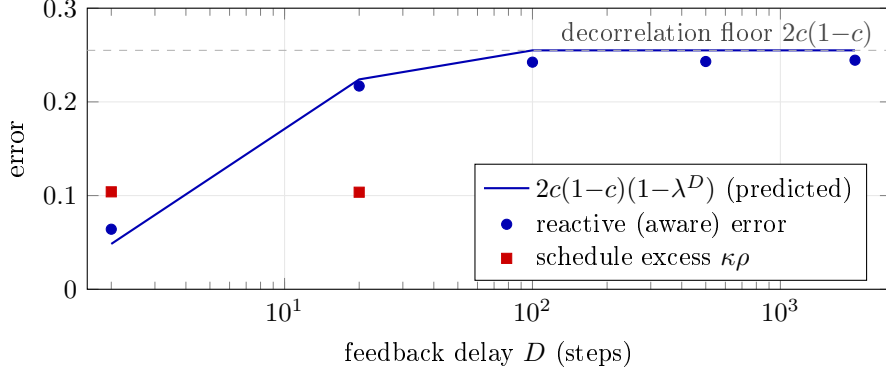


Figure 2: Delay decorrelation and schedule survival ($c = 0.15$, $\lambda = 0.90$, $\rho = 0.15$; sealed run). The reactive congestion estimate degrades along $2c(1-c)(1-\lambda^D)$ to the noise floor as feedback ages: past a modest delay it carries no information about the current state. The schedule component of the error (the $\kappa\rho$ gain from masking the known handover) is flat across a $1000\times$ span of delay — it is what survives an interplanetary round trip.

gain, and that a read-operator account explains handover-aware transport and the interplanetary collapse of feedback — not a systems claim about a specific spacecraft, the Deep Space Network, StarTCP, or Marslink. Real-trace substrates (measured DSN frames with true instrument operators; measured Starlink handover timings and loss traces) are each a separate campaign. We flag this explicitly because the discipline’s value lies in not over-claiming: the laws are derived and sealed; their transfer to fielded systems is future work.

7 Related Work

Rate-distortion theory and reverse water-filling are classical [1]; the contribution here is the consumer-relative gain in closed form and its homogeneous-consumer boundary. Handover-aware LEO transport is an active area: StarTCP proactively stalls during handovers [2], information-centric designs such as Leotp target LEO directly, and industry proposals address LEO congestion control. Our account is complementary — a read-operator model that explains *why* schedule-awareness helps and quantifies its gain — rather than a new protocol. The allocation side connects to network-utility maximization and QoS scheduling; the distinctive elements are the *derivable* margin and the blind estimation of consumer operators when they are not known in advance.

8 Conclusion

Observation Theory transfers to space communications through a single lens: distortion and reliability are consumer-relative, and the useful structure of a scarce, delayed channel is whatever the consumer’s operator can read from it. Two laws follow — the AM/GM allocation gain and the $\kappa = 1 - 2c$ / delay-decorrelation transport account — each derived a priori, sealed before measurement, and passed on disjoint seeds, with the transport coefficient machine-checked. The same framework that predicts a real gain for a deep-space multi-instrument downlink predicts none for a homogeneous broadband link, and predicts that only a deterministic schedule survives an interplanetary round trip. What the theory owes next is the move from synthetic mechanism to fielded trace.

References

- [1] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed., Wiley, 2006.
- [2] “StarTCP: A Handover-aware Transport Protocol for Starlink,” in *Proc. 8th Asia-Pacific Workshop on Networking (APNet)*, 2024.
- [3] A. H. Bond, “Observation Theory: Geometry, Distortion, and Reliability as Properties of Observation,” preprint, 2026.